Comparative Study of Deep Learning Models for Real-Time Anomaly Detection in Neonatal NICU Vital Signs

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Abstract

In Neonatal Intensive Care Units (NICUs), continuous monitoring of physiological signals is essential for early detection of critical conditions in infants. However, conventional monitoring systems often generate excessive false alarms, reducing clinical efficiency and delaying response to true emergencies. This study proposes a real-time anomaly detection framework based on deep learning applied to multivariate vital sign data, including heart rate, respiratory rate, temperature, oxygen saturation, and blood pressure. A comparative analysis of LSTM, GRU, and 1D-CNN models is conducted to evaluate their effectiveness in time-series modelling. Experimental results show that the LSTM model achieves the best performance with an accuracy of accuracy of 94.3% and a false alarm rate of 3.7%, owing to its ability to capture long-term temporal dependencies. The proposed system demonstrates strong potential for improving monitoring reliability, reducing unnecessary alerts, and supporting timely clinical interventions in NICU environments. Future work will focus on multi-modal data integration and explainable AI techniques to enhance interpret-ability.

Keywords
Neonatal Intensive Care Unit (NICU), Anomaly Detection, LSTM, GRU, 1D-CNN, Time-Series Analysis, False Alarm Reduction, Clinical Decision Support.

1. Introduction

Neonatal Intensive Care Units (NICUs) require continuous monitoring of vital physiological parameters such as heart rate, respiratory rate, blood pressure, temperature, and oxygen saturation to detect critical conditions in newborns. Even minor deviations in these signals can indicate severe complications, making timely and accurate analysis essential. However, real-time interpretation of multivariate physiological data remains challenging. Clinicians must analyze multiple interdependent signals under high-pressure conditions, increasing the risk of missing subtle patterns. Traditional threshold-based monitoring systems often fail to capture temporal relationships among vital signs, leading to excessive false alarms and reduced responsiveness. Recent advances in deep learning enable effective analysis of complex time-series data by capturing temporal dependencies and nonlinear relationships. Models such as LSTM, GRU, and CNN have shown strong performance in healthcare analytics and anomaly detection tasks. In this work, we propose a deep learning-based framework for real-time anomaly detection in NICU vital signs and perform a comparative evaluation of LSTM, GRU, and 1D-CNN models. The goal is to improve detection accuracy while minimizing false alarm rates, thereby enhancing monitoring reliability and supporting timely clinical decision-making. Deep learning has emerged as a promising solution to overcome the limitations of traditional threshold-based monitoring systems. Unlike rule-based approaches, deep learning models can automatically learn complex temporal patterns and nonlinear relationships from multivariate physiological data. This makes them particularly effective for analyzing neonatal vital signs, where subtle and gradual changes may indicate critical conditions. Among these models, Long Short-Term Memory (LSTM) networks are especially suitable due to their ability to capture long-term dependencies in sequential data, enabling more accurate and timely detection of anomalies.

1.1 Problem Statement

The dynamic and highly variable nature of neonatal physiology poses significant challenges for conventional monitoring systems. Traditional approaches based on predefined threshold values are often inadequate for detecting early signs of critical conditions such as sepsis, respiratory failure, and cardiac complications. In NICU environments, vital signs—including heart rate, respiratory rate, and oxygen saturation—are strongly interdependent; however, many systems analyze them independently, limiting their ability to capture meaningful relationships. As a result, these systems generate excessive false alarms while failing to identify subtle early warning patterns. Additionally, threshold-based methods rely on fixed limits that do not account for patient-specific variability or evolving clinical conditions. Consequently, gradual but clinically significant changes may go unnoticed, delaying timely intervention in high-risk neonatal care settings. To address these limitations, this study proposes a deep learning-based framework for real-time analysis of multivariate NICU time-series data, enabling improved anomaly detection by capturing both temporal dependencies and interrelationships among physiological signals.

1.2 Objectives

A. To model temporal patterns in vital sign data by analyzing interactions among heart rate, respiratory rate, oxygen saturation, and temperature.
B. To detect early signs of physiological deterioration through identification of abnormal patterns and deviations.

C. To balance detection accuracy and reliability by minimizing false alarms while maintaining high sensitivity.

D. To develop a real-time framework for continuous monitoring and timely alert generation.
E. To compare LSTM, GRU, and 1D-CNN models using performance metrics such as accuracy, precision, recall, and false alarm rate.
F. To support clinical decision-making by providing reliable and interpretable outputs.

2. Literature Review

Deep learning has significantly improved the analysis of physiological data in healthcare, particularly for continuous monitoring applications. Traditional anomaly detection methods, including statistical and rule-based approaches, often struggle with the complexity of NICU data, which is multivariate, nonlinear, and highly dynamic. These limitations lead to excessive false alarms and reduced sensitivity to subtle physiological changes. In contrast, deep learning models such as LSTM, GRU, and CNN effectively capture temporal dependencies and nonlinear relationships among vital signs, enabling more accurate anomaly detection. LSTM networks are particularly well-suited for modeling sequential data and identifying both abrupt anomalies and gradual trends in neonatal physiological signals. Recent studies report that deep learning approaches outperform conventional methods in detecting clinically relevant patterns and improving prediction accuracy. Furthermore, their integration into real-time monitoring systems and clinical decision support systems enhances early risk detection, reduces false alarm rates, and supports timely clinical interventions in NICU environments.

3. Methodology

This section presents the methodology for developing the proposed real-time anomaly detection system for NICU monitoring. The framework utilizes multivariate time-series data consisting of key physiological parameters, including heart rate (HR), respiratory rate (RR), oxygen saturation (SpO_2), and body temperature. To capture temporal dependencies and interrelationships among these signals, a deep learning approach based on Long Short-Term Memory (LSTM) networks is employed. The model is designed to learn both short-term variations and long-term trends associated with early stages of clinical deterioration. The overall framework follows a structured pipeline, where raw physiological data undergo preprocessing steps such as normalization, noise filtering, and sequence windowing. The processed data is then used to train the model to distinguish between normal and anomalous patterns. Model performance is evaluated using standard metrics, including accuracy, precision, recall, and false alarm rate. Additionally, the system supports near real-time inference, enabling continuous monitoring and timely alert generation.

3.1 Data Collection

A publicly available NICU dataset is used, consisting of time-stamped physiological recordings of neonatal vital signs. These continuous measurements enable analysis of temporal patterns and trends in patient health.

The dataset includes the following key features:

- Heart Rate (HR) – measured in beats per minute (bpm)
- Respiratory Rate (RR) – measured in breaths per minute
- Oxygen Saturation (SpO_2) – expressed as a percentage
- Body Temperature – measured in degrees Celsius

The dataset is preprocessed using min–max normalization to scale each feature to a common range:

$$x_{i,j}^{(norm)} = \frac{x_{i,j} - x_{min,j}}{x_{max,j} - x_{min,j}}$$

In addition, missing and erroneous values are handled using cleaning and imputation techniques. Temporal gaps in the data are addressed through interpolation to maintain continuity. These preprocessing steps improve data quality and ensure suitability for training deep learning models in a real-time NICU monitoring framework.

3.2 Model Architecture

Long Short-Term Memory (LSTM) networks are employed in this study due to their ability to capture long-range temporal dependencies in sequential data. In NICU monitoring, physiological signals exhibit temporal continuity and interdependence, making LSTM models well-suited for detecting both abrupt and gradual anomalies. The proposed model processes multivariate time-series data using fixed-length sliding windows and outputs a binary classification indicating normal or anomalous behavior. The architecture consists of stacked LSTM layers followed by a dense output layer for final prediction.

3.3 Training and Validation

3.3.1 Data Splitting

The dataset is divided into training (70%), validation (15%), and test (15%) sets. The training set is used for model learning, while the validation set is used for hyperparameter tuning. The test set is used to evaluate final model performance.

3.3.2 Loss Function

Binary cross-entropy loss is used to measure the difference between predicted and actual labels.

3.3.3 Optimizer

The Adam optimizer is used for model training due to its efficiency and fast convergence.

Table 1: Sample NICU Dataset

Time Stamp	Heart Rate (bpm)	Respiratory Rate (breaths/min)	SpO_2 (%)	Temperature (°C)
00:00:01	138	36	98	36.6
00:00:02	140	35	97	36.7
00:00:03	142	34	98	36.8
00:00:04	141	36	99	36.9
00:00:05	143	37	97	37.0
00:00:06	145	38	96	37.1
00:00:07	144	37	97	37.0
00:00:08	146	39	96	37.2

3.4 Overall System Architecture

The proposed system is designed for real-time anomaly detection in NICU environments using multivariate physiological time-series data. The framework follows a structured pipeline consisting of data acquisition, preprocessing, temporal modelling, and alert generation, as illustrated in Fig. 1. Physiological signals such as heart rate, respiratory rate, oxygen saturation, and body temperature are continuously collected from bedside monitoring devices. These signals are preprocessed using normalization, noise filtering, and sequence windowing to ensure data consistency and quality. The processed data is then fed into a stacked LSTM-based model, which captures temporal dependencies and interrelationships among vital signs to identify both abrupt and gradual anomalies. The model outputs a binary classification indicating normal or anomalous behavior. Finally, the system operates in a near real-time environment, where incoming data streams are continuously analyzed. Detected anomalies trigger alert notifications, enabling timely clinical intervention and reducing the risk of delayed response.

Fig. 1: End-to-End Framework for Real-Time Multivariate Time-Series Anomaly Detection in NICU

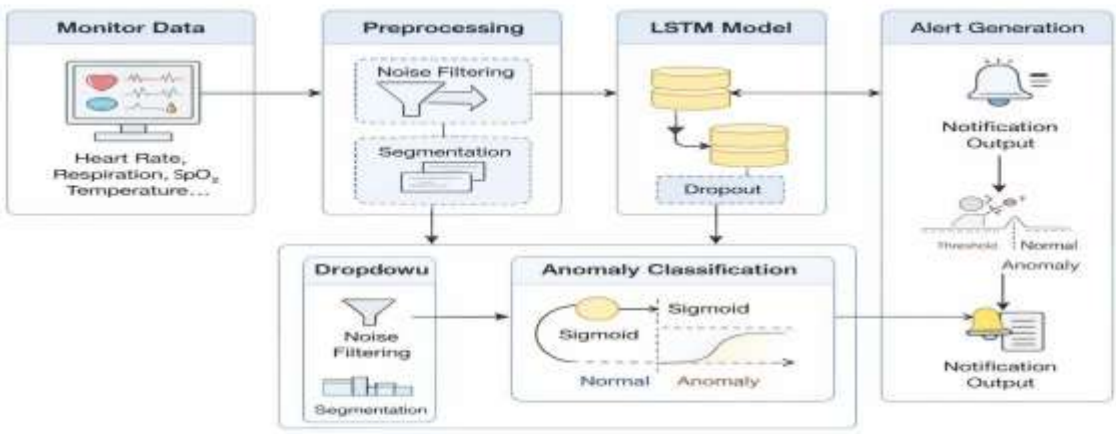


Fig. 1: Overall System Architecture for NICU Anomaly Detection

A sample of processed physiological data with anomaly labels is shown in Table 2.

Time stamp	Heart Rate (bpm)	Respiratory Rate (breaths/min)	SpO_2 (%)	Temperature (°C)	Anomaly (0/1)
00:00:01	138	36	98	36.6	0
00:00:02	140	35	97	36.7	0
00:00:03	144	37	96	36.9	1
00:00:04	149	39	95	37.1	1
00:00:05	152	41	94	37.3	1
00:00:06	155	42	93	37.4	1
00:00:07	148	38	95	37.0	0

3.4 Real-Time Monitoring and Anomaly Detection

Once trained, the model is deployed in a real-time monitoring framework for continuous analysis of neonatal vital signs. Streaming physiological data are processed using fixed-length sliding windows to maintain temporal consistency. For each input sequence, the model outputs an anomaly score representing the likelihood of abnormal behavior. An alert is triggered when the predicted score exceeds a predefined threshold. This threshold can be adjusted to balance sensitivity and specificity, helping reduce false alarms while ensuring timely detection of critical conditions. This makes the system suitable for high-acuity NICU environments.

3.5 Sample Data for Real-Time Analysis

The real-time input consists of continuously streamed multivariate vital sign data, as shown in Table 2. Each record includes time-stamped measurements of heart rate, respiratory rate, oxygen saturation, and body temperature. The dataset contains both normal and abnormal patterns, including gradual deviations and sudden changes in physiological signals. The model analyses these sequential inputs to detect clinically significant variations and generate alerts when anomalies are identified, supporting timely clinical intervention.

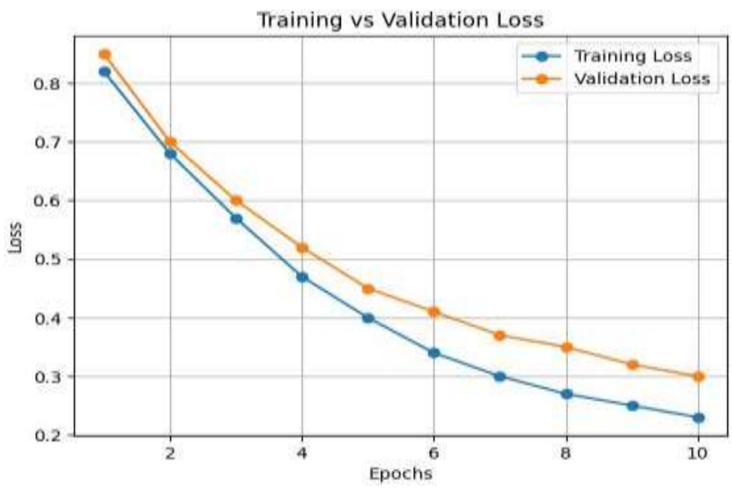
4. Experiments and Results

This section presents the evaluation of the proposed anomaly detection framework. Three models—LSTM, GRU, and 1D-CNN—are compared using the same dataset and training pipeline to ensure a fair comparison. All models are trained using the Adam optimizer and binary cross-entropy loss. Performance is evaluated using accuracy, precision, recall, F1-score, and false alarm rate, along with sensitivity and specificity to assess clinical reliability. Results indicate that the LSTM model outperforms GRU and 1D-CNN in capturing temporal dependencies, achieving higher detection accuracy and lower false alarm rates.

4.1 Experimental Setup

The models are implemented using Python with TensorFlow and Keras. The dataset consists of approximately 62,500 multivariate NICU records, including heart rate, respiratory rate, oxygen saturation, and body temperature. Data preprocessing includes noise filtering, normalization, missing value handling, and segmentation into fixed-length time windows to preserve temporal context. The dataset is split into training, validation, and test sets for model development and evaluation. All models are trained for 50 epochs with a batch size of 64 using the Adam optimizer. Hyperparameters such as learning rate, hidden units, and window size are tuned using validation performance. Early stopping is applied to prevent overfitting and improve generalization.

The training behavior of the model is illustrated in Fig. 2.



The comparative performance of the models is summarized in Table 3.

Models	*Accuracy (%)*	*Precision (%)*	*Recall (Sensitivity) (%)*	*F1-Score (%)*
LSTM	*94.3*	*93.5*	*91.2*	*92.3*
GRU	*93.7*	*92.4*	*90.1*	*91.2*
1D-CNN	*92.8*	*91.3*	*89.2*	*90.2*

4.2 Performance Metrics

Model performance is evaluated using standard classification metrics. Accuracy represents the proportion of correctly classified instances, while precision measures the correctness of anomaly predictions. Recall indicates the ability to detect actual anomalies, and F1-score provides a balanced measure of precision and recall. These metrics collectively assess the effectiveness of each model in identifying anomalies while minimizing errors.

4.2.1 Training Behavior Analysis

Figure 2 shows the variation of training and validation loss across epochs. Both curves exhibit a consistent downward trend, indicating effective learning. The small gap between training and validation loss suggests stable convergence and minimal overfitting, demonstrating good generalization to unseen data.

The false alarm rate comparison is presented in Table 4

Model	**False Alarm Rate (%)**
LSTM	**3.1**
GRU	**3.9**
1D-CNN	**4.8**

4.2.2 False Alarm Analysis

As shown in Table 4, the LSTM model achieves the lowest false alarm rate compared to GRU and 1D-CNN. This indicates superior reliability in distinguishing true anomalies from normal physiological variations. Reducing false alarms is critical in NICU environments, as excessive alerts can lead to alarm fatigue and delayed clinical response.

4.3 Results and Comparisons

4.3.1 Model Comparison

Table 3 summarizes the performance of LSTM, GRU, and 1D-CNN models across accuracy, precision, recall, and F1-score. The LSTM model consistently achieves the highest values across all metrics, demonstrating its effectiveness in capturing temporal dependencies in multivariate physiological data. Figure 3 provides a visual comparison of model performance, where LSTM maintains superior and stable results across all metrics. In contrast, GRU and 1D-CNN show comparatively lower performance, indicating limitations in modelling long-term temporal patterns. Models with balanced performance across multiple metrics are more suitable for clinical applications, where both accuracy and reliability are critical.

Fig. 3: Comparative Analysis of Model Performance with Highlighted Best Model (LSTM)

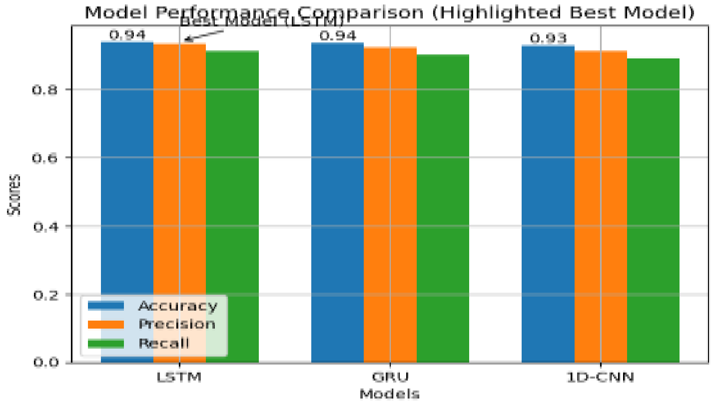


Figure 3 highlights the superior performance of the LSTM model across all evaluation metrics. Its ability to capture long-term temporal dependencies results in higher accuracy and more reliable anomaly detection compared to GRU and 1D-CNN models.

Table 5: Final Training and Validation Loss after 50 Epochs
The training and validation loss values for all models are presented in Table 5.

Model	Training Loss	Validation Loss
LSTM	**0.17**	**0.20**
GRU	**0.19**	**0.23**
1D-CNN	**0.21**	**0.25**

Table 5 shows that the LSTM model achieves the lowest training and validation loss, indicating better convergence and generalization. The smaller gap between training and validation loss further suggests reduced overfitting and improved model stability compared to GRU and 1D-CNN.

Fig. 4: (a) Precision–Recall Curve and (b) ROC Curve for Model Performance Evaluation

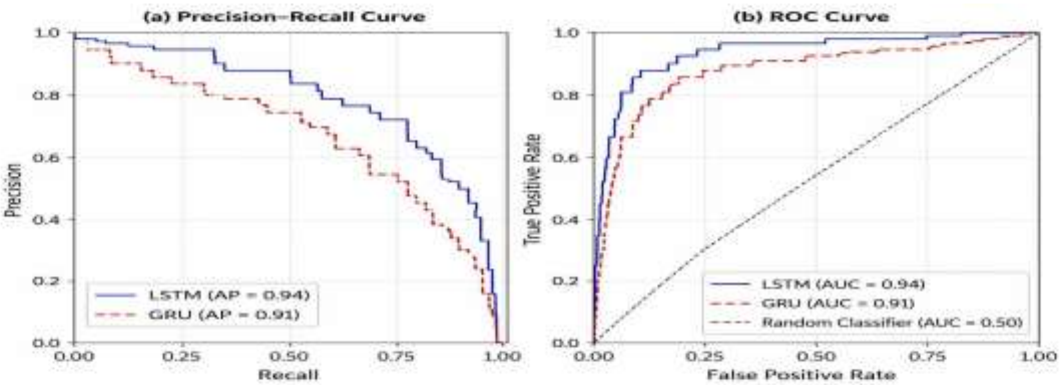


Figure 4 presents the Precision–Recall and ROC curves for the LSTM model. The Precision–Recall curve highlights the model's effectiveness in handling class imbalance, while the ROC curve demonstrates its ability to distinguish between normal and anomalous instances. The consistent performance across both curves indicates strong discriminative capability.

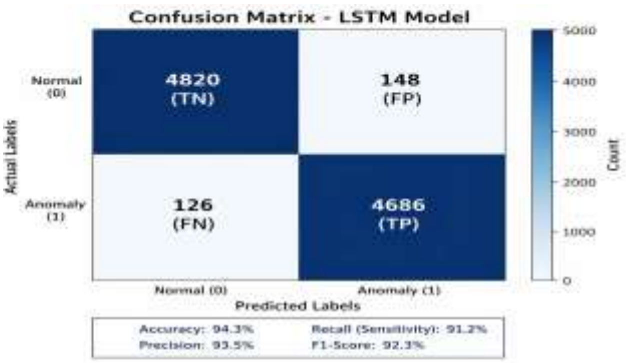
4.4 Visualization of Results

To further evaluate model performance, Precision–Recall and ROC curves are generated for the LSTM model. The Precision–Recall curve illustrates the trade-off between correctly detected anomalies and false positives, which is particularly important in imbalanced datasets. The ROC curve represents the relationship between true positive rate and false positive rate across different threshold values, providing insight into the model's classification capability.

4.5 Discussion of Results

The results show that the LSTM model consistently outperforms GRU and 1D-CNN across all evaluation metrics, achieving an accuracy of 94.3%, precision of 93.5%, recall of 91.2%, and an F1-score of 92.3%. This performance highlights the ability of LSTM networks to capture long-term temporal dependencies in multivariate physiological signals, which is essential for early detection of clinical deterioration. In terms of false alarm rate, the LSTM model achieves the lowest value (3.1%), compared to GRU (3.9%) and 1D-CNN (4.8%). This reduction is critical in NICU environments, where excessive alarms can lead to alarm fatigue and delayed clinical response. The improved precision and lower false positive rate demonstrate the reliability of the LSTM-based system in real-world scenarios. From a training perspective, the LSTM model shows stable convergence, with training and validation loss stabilizing around 25–30 epochs and reaching final values of 0.17 and 0.20, respectively. The small gap between these values indicates minimal overfitting and strong generalization. In contrast, GRU and 1D-CNN exhibit higher validation loss and larger gaps, indicating comparatively weaker performance. Furthermore, ROC analysis shows that the LSTM model achieves a higher Area Under the Curve (AUC ≈ 0.93–0.95), reflecting superior discriminative ability between normal and anomalous states. Overall, the LSTM-based framework provides a strong balance between accuracy, sensitivity, and false alarm reduction, making it well-suited for real-time NICU monitoring.

Fig. 5: Confusion Matrix of the LSTM Model



5 Discussion

This study presents a real-time anomaly detection framework for NICU monitoring using deep learning models. A comparative evaluation of LSTM, GRU, and 1D-CNN demonstrates that the LSTM model consistently achieves superior performance across all evaluation metrics. The improved performance of LSTM is attributed to its ability to capture long-term temporal dependencies and complex relationships in multivariate physiological signals. This is particularly important in NICU environments, where anomalies often develop gradually rather than abruptly. From a clinical perspective, the lower false alarm rate achieved by the LSTM model significantly reduces alarm fatigue and improves the reliability of monitoring systems. This enables healthcare providers to respond more effectively to critical events while minimizing unnecessary alerts. Despite these promising results, certain limitations remain. The study relies on retrospective data, and real-time deployment may introduce challenges such as sensor noise, missing values, and variability across patients. Additionally, the model lacks interpretability, which may limit clinical trust. Future work will focus on integrating multi-modal data sources and incorporating explainable AI techniques to improve transparency. Further validation in real-world NICU environments is also necessary to ensure practical applicability.

6 Conclusion

This work presents a deep learning-based approach for real-time monitoring and anomaly detection in neonatal vital sign data within NICU environments. A comparative evaluation of LSTM, GRU, and 1D-CNN models shows that LSTM delivers the most reliable performance across all metrics. The model achieves an accuracy of 94.3%, precision of 93.5%, recall of 91.2%, and a low false alarm rate of 3.1%, demonstrating its effectiveness in identifying clinically relevant anomalies. The superior performance of the LSTM model is largely due to its ability to capture temporal dependencies in sequential physiological signals. This capability is essential in NICU settings, where subtle and gradual changes in vital signs may indicate the onset of critical conditions. By reducing false alarms while maintaining high detection sensitivity, the proposed system supports more dependable monitoring and assists clinicians in making timely decisions. However, certain limitations should be considered. The current study is based on a single dataset, which may restrict generalizability across different clinical settings. In addition, the model lacks interpretability, which can limit its acceptance in real-world healthcare applications. Future work will focus on evaluating the framework on larger and more diverse datasets, as well as incorporating explainable AI techniques to improve transparency. Extending the model toward condition-specific predictions, such as early detection of sepsis or respiratory complications, is another important direction. Overall, the results highlight the practical potential of deep learning methods for improving neonatal monitoring systems. Integrating such intelligent frameworks into NICU workflows can enhance early detection, reduce alarm burden, and ultimately contribute to better patient care.

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